

So, after SILENCE and THE NOTE and INTERVALS,
what are the next most complex PITCH PATTERNS?

LESSON 14

④ THE MUST KNOW SCALES

A SCALE IS A SEQUENCE OF NOTES SEPARATED BY SPECIFIC INTERVALS.

The intervals that generally separate the notes in scales are **2nds**.

MINOR 2nds=HALF STEPS

MAJOR 2nds=WHOLE STEPS

AUGMENTED 2nds= 1 and 1/2 STEPS

SCALES are considered to be some of the most important PITCH PATTERNS in music particularly when considering the area of **MELODY**. What is a melody? The melody of the song is that familiar, LINEAR string of INDIVIDUAL NOTES most often SUNG BY THE SINGER. It's that part of the song that you find yourself humming or whistling. It is often considered the HORIZONTAL aspect of music. (HARMONY: how multiple notes sound when played SIMULTANEOUSLY, as in CHORDS, is considered the VERTICAL aspect.) In INSTRUMENTAL MUSIC, of course the melody will be played on an instrument. SOLOS are IMPROVISED MELODIES! Since melodies are thought to be based on scales, those of you interested in JAMMING and IMPROVISING will want to take particular care in learning your scales. There are dozens of scales in the Western World and indeed all over the planet! We will at first focus on what I call the MUST KNOW SCALES, the ones we use everyday in Western popular music. AND AS ALWAYS ...

YOU NEED TO KNOW YOUR SCALES ALL 4 WAYS ...

THINK

There is a lot to KNOW about scales. You have to MEMORIZE their SPELLINGS and their INTERVAL PATTERNS. You have to understand how they are used to form MELODIES and how they interact with CHORDS in songs.

READ

You need to be able to recognize them when you see them written on paper in both STANDARD NOTATION and TABLATURE. You should know how MODES relate to KEY SIGNATURES.

HEAR

Again and most importantly, you need to know what these scales SOUND LIKE. This is the whole thing when it comes to IMPROVISING. This will take lots of work but the rewards are incredible. You can pretty much do anything you want with music.

SPEAK

Finally, you have to master the myriad FINGERINGS for these scales. SCOTTY'S COLOR CODED FINGER CHARTS are awesome tools to use and the MODE JAM TAPE will make it much more fun. But still you have to commit yourself to some DISCIPLINED "grunt work".

THE MUST KNOW SCALES

CHROMATIC

1	#1 b2	2	#2 b3	3	4	#4 b5	5	#5 b6	6	#6 b7	7	8
---	----------	---	----------	---	---	----------	---	----------	---	----------	---	---

MAJOR DIATONIC IONIAN MODE

1	2	3	4	5	6	7	8	I
---	---	---	---	---	---	---	---	---

LYDIAN

1	2	3	#4	5	6	7	8	IV
---	---	---	----	---	---	---	---	----

MIXOLYDIAN

1	2	3	4	5	6	b7	8	V
---	---	---	---	---	---	----	---	---

MELODIC MINOR

1	2	b3	4	5	6	7	8
---	---	----	---	---	---	---	---

← b6 b7 →

DORIAN

1	2	b3	4	5	6	b7	8	ii
---	---	----	---	---	---	----	---	----

PHRYGIAN

1	b2	b3	4	5	b6	b7	8	iii
---	----	----	---	---	----	----	---	-----

AEOLIAN MODE NATURAL MINOR RELATIVE MINOR

1	2	b3	4	5	b6	b7	8	Vi
---	---	----	---	---	----	----	---	----

HARMONIC MINOR

1	2	b3	4	5	b6	7	8
---	---	----	---	---	----	---	---

MAJOR PENTATONIC

1	2	3	5	6	8	I
---	---	---	---	---	---	---

MINOR PENTATONIC

1	b3	4	5	b7	8	Vi
---	----	---	---	----	---	----

WHOLE TONE

1	2	3	#4	#5	#6	8	+
---	---	---	----	----	----	---	---

DIMINISHED

1	2	b3	4	b5	b6	bb7	7	8	O
---	---	----	---	----	----	-----	---	---	---

LOCRIAN

1	b2	b3	4	b5	b6	b7	8	Vii°
---	----	----	---	----	----	----	---	------

MORE ABOUT SCALES

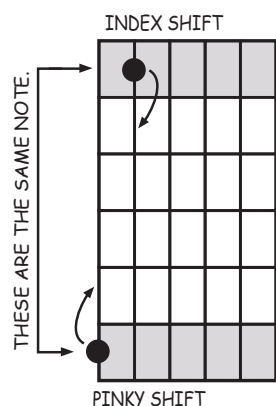
LESSON 15

You probably already know that one big thing about scales is that you have to ultimately know how to PLAY them on the neck of your guitar. All scales have MANY FINGERINGS because you can play any one in any of the 12 keys anywhere on the fret board. You don't need to know all the fingerings, but over your career, you owe it to yourself to soak up as many of them as you can. Your COLOR CODED FINGER CHARTS show 12 ways of playing each scale. These are the possible ways to play the scale within our 6 fret CHROMATIC "BOX" orientation. There are indeed other ways to play these scales as well (many involving SLIDES from one box to another, and you will pick these up over time from various sources). Of course, you could play any given scale on a SINGLE STRING simply by proceeding up the string using the correct pattern of half and whole steps for that scale. But how PRACTICAL a use of your fingers is that? What you're always trying to do is make the BEST and MOST LOGICAL use of all your fingers! That is the only way to attain real speed and control. Many players do not understand these 12 basic fingerings simply because they do not really understand their fret board in the way that YOU DO! You know that you can play all of the 12 notes, within a 5 fret range, for a distance of 2 and 1/3rd octaves anywhere on the neck! You understand how INDEX and PINKY SHIFTS work. What this means is that you should be able to play ANY SCALE... IN ANY KEY... ANYWHERE ON THE NECK! It may be a while before you see how AWESOME it is that you know this stuff and how far ahead of other players this puts you!

Many of these suggested guidelines for scale fingerings may at first seem more difficult or awkward than what you might come up with. But that is often simply because you haven't worked with them enough yet. Your fingers and ligaments haven't stretched enough yet to get the full benefit. You don't have enough EXPERIENCE to see the theoretical significance they hold. A classic example would be all those players who don't use their PINKY because it's such a weak finger. But it's only weak because you DON'T USE IT! You only have 4 fingers to begin with! Don't sacrifice your pinky just because you're too LAZY to get in there and work with it! That's 25% of your hand! It's a lot of work mastering these fingerings, but at the end comes TOTAL FREEDOM of a sort most players never even see is there! Your MODE JAM TAPE will make it more fun to practice your MAJOR DIATONIC MODES and your PENTATONIC SCALES. These are the most common scales and so the first ones you will want to get working on. But you will ultimately have to commit to a FOCUSED and DISCIPLINED practice regimen to master all your scales.

But first we must make sure that we know how to finger scales in general. We have to go back to what we learned about fingering CHROMATIC SCALES.

THE KEY TO UNDERSTANDING SCOTTY'S FAMOUS COLOR CODED FINGER CHARTS



Hopefully you all remember the significance of the 6 FRET CHROMATIC BOX. It takes 5 FRETS to control the 12 notes on the fret board, but that 5th note can be played with either a PINKY SHIFT on one string or an INDEX SHIFT on the next higher string. These can be any 6 frets anywhere on the neck. Remember that your G and B strings are tuned one fret closer together and that is bound to effect your fingering. As a rule, you'll find yourself using index shifts when going up the scale and pinky shifts when coming down but you will see that that isn't always the logical thing to do. Often, using INDEX REACHES when going both up and down help you avoid pinky shifts. When coming back onto the white frets after a FINGER REACH, make sure you use the correct finger that normally would have controlled that fret.

SO WHERE IS THE ROOT NOTE OF THE SCALE?

The root note can be any one of the 12 notes. That's how you tell which key you are playing the scale in. So the root can be under any finger on any fret on any string. It can even be a finger shift note on the shaded frets. That's what accounts for all these different finger patterns for the same scale. Note that as you go from box to box on any one of your charts, the RED DOT which is the ROOT NOTE, moves up one fret each time and so you are playing in the next higher key.

SO HOW DOES THE COLOR CODING SYSTEM WORK?

Remember back in grade school science class you learned the 7 COLORS OF THE SPECTRUM? They go

RED - ORANGE - YELLOW - GREEN - BLUE - INDIGO - VIOLET

Notice how most of our most common scales contain 7 NOTES (not including the octave). These are called the DEGREES OF THE SCALE. In our charts, each of the degrees of the scale is represented by each of the corresponding colors in the rainbow. So your ROOT NOTE is RED ...your 2nd is ORANGE ...your 3rd is YELLOW ...your 4th is GREEN ...your 5th is BLUE ...your 6th is INDIGO ...and your 7th is VIOLET. If for example, your 3rd is flatted (as in a minor scale), your YELLOW dot will have a flat sign on it etc. These color coded charts are so awesome because you can see all the degrees of the scale at a glance once you've memorized the color sequence. Compare that to your boring old black and white charts!

SO A MAJOR DIATONIC SCALE WOULD GO...



AND A NATURAL MINOR SCALE WOULD GO...



AND, SO INTERESTINGLY, THE TRIADS ARE IN PRIMARY COLORS!



STILL MORE ABOUT SCALES

LESSON 16

As we said before, the subject of scales is perhaps the most substantial subject in music theory. As always, we need to know these pitch patterns **4 WAYS ...THINK - READ - HEAR - SPEAK**. In this lesson, we look deeper into the **INTERVALS** that separate the notes in scales and begin to listen to them. Eventually you will want to **MEMORIZE** the sound of these scales so you can use them in composition and improvisation.

Many people are shocked to find out how many scales there are ...and all these weird names like **MIXOLYDIAN** and such. You might have taken quite a lot of music lessons and barely heard of them. Most of us have only heard of the **MAJOR SCALE**. Did you ever understand a **CHROMATIC SCALE** before you took this program? You might have heard that **PENTATONIC SCALES** and **BLUES SCALES** are big in Rock and Roll. But that's about it. How can Scotty suggest that all these other scales are equally important to know and even equally common in the songs we play? The first thing to know is this ...

THERE IS A SEPARATE SCALE FOR EACH CHORD IN THE SONG.

It is often suggested that an entire song is based on one scale (and certainly you can look at music that way but you might miss some important revelations). In Western Music, that fundamental scale is most often said to be the **MAJOR DIATONIC SCALE**. That's why it was the first one they tried to teach us back in grade school. A more modern way of teaching theory suggests that it is more useful to consider a separate scale for each chord in the song. How many chords are in the average pop song? Maybe between 3 and 6 ...sometimes more. This means that you are playing 3 to 6 of these scales in every tune you play! We most often see a mixture of **MAJOR** and **MINOR** chords in songs. What good is it to view the song as only containing one scale (say the **MAJOR SCALE**) if you have both major and minor chords? Doesn't it stand to reason that minor scales would work with minor chords? You will find that indeed there are very specific guidelines as to what scale goes with what chord. How do we know which scale works with which chord? This is a complex issue, but first consider this ...

THERE IS A CHORD EMBEDDED INSIDE EVERY SCALE
AND THE SCALE THAT GOES WITH THAT CHORD IS THE SCALE THAT HAS THAT CHORD INSIDE IT!
(Hopefully by this time you remember the number spellings for your major and minor chords)

MAJOR 1-3-5

MINOR 1-b3-5

In the video, we use **SILENT NIGHT** to demonstrate that even these strange scales like **LYDIAN** and **MIXOLYDIAN** show up in the most normal of songs if that is what the chord progression dictates. We will be looking at these relationships more in depth in our next few lessons on **MODES** and **HARMONIZATION**.

SO WHAT DO THESE SCALES SOUND LIKE?

CHROMATIC

Hopefully by now you are becoming familiar with the sound of the CHROMATIC SCALE. It is 13 notes separated by 12 half steps. You don't really play the complete scale very often in pop music but a lot of songs do contain brief chromatic passages. You need to know what a half step sounds like.

MAJOR SCALES

We've been listening to the MAJOR DIATONIC SCALE since we were kids. Now we are learning that there is MORE THAN ONE MAJOR SCALE. A major scale has a major chord inside it ...**1-3-5** and there are several scale patterns that fit that category. The #4 in LYDIAN may sound strange at first but you will notice it more and more as your ears become accustomed to it. Indeed it often "hides" in the melody to where you didn't even notice it there before. You will come to recognize the b7 in MIXOLYDIAN as being an INCREDIBLY common and important note in many contexts.

MINOR SCALES

There are MINOR CHORDS INSIDE MINOR SCALES ...**1-b3-5**. Most people would agree that the MINOR 3rd makes things sound SAD. Once again there are several slightly different minor scale patterns ...many again containing that b7. You may find it hard to believe that you will ever be able to pick up on their subtle differences by ear but indeed you can and must. Many pieces of music are written in MINOR KEYS where these scales replace the MAJOR SCALE as the central scale. In songs, minor scales interact with minor chords in very specific ways. Notice that the HARMONIC MINOR SCALE contains an AUGMENTED 2nd (1 and 1/2 step) interval.

PENTATONIC SCALES

PENTATONIC MEANS 5 NOTES. These scales contain two fewer notes per octave than their 7 note DIATONIC cousins. Therefore, these scales contain TWO AUGMENTED 2nd intervals which give them an open and "airy" sound that has attracted a lot of attention by various cultures throughout history. Some of you have heard that these are the ONLY scales you need to know to play Rock and Roll and Blues music but this is a terrible over simplification. Why only try to control 5 notes per octave when there is so much more? The MINOR PENTATONIC SCALE is often thought of as the foundation for the BLUES SCALE. It does contain the MINOR 3rd giving it a sad or "bluesy" feel. The b7 is also a huge note in blues. Try to remember what they sound like.

SYMMETRICAL SCALES

SYMMETRICAL SCALES GO WITH SYMMETRICAL CHORDS like AUGMENTED and DIMINISHED. A WHOLE TONE SCALE is all whole steps. A CHROMATIC SCALE is all half steps. A DIMINISHED SCALE goes whole-half-whole-half etc. Notice that these scales DO NOT CONTAIN NOTE 5 so they are NEITHER MAJOR OR MINOR! There are far fewer of these scales and chords in popular music partially because we are so used to that all important 5th that we don't quite know what to do when it isn't there. It is harder for us to memorize what these patterns sound like. Symmetrical chords are much more common in Jazz and Classical music, often snuck in between major and minor chords. LOCRIAN also has a b5 but notice that it is not symmetrical.

DIATONIC HARMONY A LA MODE

LESSON 17

WHY DO WE SEE THE SAME SETS OF CHORDS OVER AND OVER AGAIN IN DIFFERENT SONGS?

Those of you who have been playing for awhile will agree that there are hundreds of pop tunes that all share the same set of chords. One of the most common combinations in Pop Rock Music certainly is

G MAJOR - C MAJOR - D MAJOR - E MINOR - A MINOR

You might have played dozens of songs that have some or all of these chords in them. Why is that? Well you could just say that perhaps most song writers think that these chords sound good together. But there is a deeper level on which these chords are connected. Consider this statement ...

MOST SONGS CONTAIN ONLY 7 OF THE 12 NOTES. WHICH 7? THE 7 NOTES IN A MAJOR DIATONIC SCALE IN A GIVEN KEY.

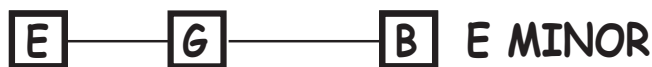
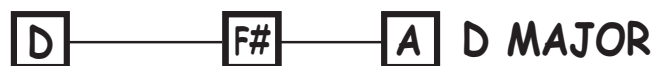
It's our good old friend the MAJOR DIATONIC SCALE again. So much of Western Music can be traced back to it. Now we find that the average song contains only the 7 notes in this scale. Not only is the MELODY of the song constructed out of these notes,

THE SONG CONTAINS ONLY THE CHORDS THAT CAN BE CONSTRUCTED OUT OF THOSE 7 NOTES!

Let's look at an example in the KEY OF G. The 7 notes in a G MAJOR SCALE are



WHAT CHORDS CAN BE MADE OUT OF THESE 7 NOTES?



AS YOU CAN SEE, THIS SET INCLUDES ALL THE CHORDS IN OUR EXAMPLE!

SO WHAT'S THIS GOT TO DO WITH MODES?

This is HUGE. You've learned a lot about PITCH PATTERNS so far in this program. We started with SILENCE and then we learned that all conventional patterns in music have a ROOT NOTE. We then talked about INTERVALS and suggested that the true nature of these patterns involves the SPACING of the notes. Now we're looking at the more complex note patterns like SCALES. You know all this stuff but you still don't see how this all comes together to EXPLAIN the songs we play. Well we are about to look at that now. We're going to do it in a rather unique way involving the concept of MODES. Many people have heard of modes. In fact many music students are taught a great deal about them without ever being shown their significance and relevance to the subject of how SCALES and MELODIES relate to CHORD PROGRESSIONS in conventional songs. The question is ...

HOW DO YOU KNOW WHAT SCALE GOES WITH WHAT CHORD?

Many of us who have done some JAMMING still remember when someone first told us that a single scale (often the MAJOR SCALE or the PENTATONIC SCALE) could be used to create melodies over a given set of chords. Wow! It seemed to make improvising so easy! We sound halfway like ERIC CLAPTON all ready! But if you play this way for long you begin to run out of ideas. You notice that all the notes in the scale sound pretty good but any given note will sound better or worse depending on which chord you play it over. What's going on? What do the guys on the radio know that you don't?

WOW! I CAN PLAY ONE SCALE OVER SEVERAL CHORDS!

NO! WHEN THE CHORD CHANGES, SO MUST THE SCALE CHANGE!

There is a separate scale for each chord. This makes sense doesn't it? How, for example, could a MAJOR SCALE work over both major and minor chords? Shouldn't we be playing minor scales over the minor chords? But it seems that the majority of people we talk to have never considered this possibility. They point out that playing one scale sounds pretty good and it would be too hard to switch back and forth from scale to scale every time the chord changes. Who ever heard of such a thing? But there is an answer.

THE SAME SET OF NOTES IS SEVERAL DIFFERENT SCALES. THE QUESTION IS WHICH NOTE IN THE SET IS "DO"?

THIS IS THE WHOLE THING ABOUT MODES.

MORE ABOUT MODES

LESSON 18

THE SAME SET OF NOTES IS SEVERAL DIFFERENT SCALES.

This is the whole concept of MODES. But what does this mean and why is it important for you to understand about them? Let's look at 2 octaves of a MAJOR DIATONIC SCALE. We recognize it by it's familiar pattern of intervals ... mostly WHOLE STEPS except between **3 and 4** and **7 and 8** or, in other words ...

WHOLE - WHOLE - HALF - WHOLE - WHOLE - WHOLE - HALF

But if you start on one of the other notes in the set, the interval pattern shifts and it is no longer accurate to view the new sequence of intervals as a Major Diatonic Scale. It is one of the MODES of the Major Diatonic Scale.

WHOLE	WHOLE	HALF	WHOLE	WHOLE	WHOLE	HALF	WHOLE	WHOLE	HALF	WHOLE	WHOLE	HALF
1	2	3 4	5	6	7 8	This 1st mode is the IONIAN ...the MAJOR DIATONIC SCALE itself. Notice that it is a MAJOR SCALE with a MAJOR CHORD (1-3-5) embedded inside it.						

1	2 b3	4	5	6 b7	8	Start on the 2nd note and your interval pattern changes. This 2nd mode, DORIAN is a MINOR SCALE containing a MINOR CHORD (1-b3-5)
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1 b2	b3	4	5 b6	b7	8	Start on the 3rd note and you're playing PHRYGIAN MODE . It is a MINOR SCALE. Notice the b2 and b6.
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1	2	3	#4 5	6	7 8	Start on the 4th note and you're playing LYDIAN MODE . It is a MAJOR SCALE with a #4.
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Start on the 5th note and you are playing the **MIXOLYDIAN**, a very important MAJOR SCALE with a b7.

1	2	3 4	5	6 b7	8
---	---	-----	---	------	---

This 6th mode is the **AEOLIAN**. Notice that it is a NATURAL MINOR SCALE. It is the very important RELATIVE MINOR mode.

1	2 b3	4	5 b6	b7	8
---	------	---	------	----	---

Start on the 7th note and you're playing the **LOCRIAN MODE**. It is the only mode that is neither major or minor. It's b5 and b3 relates it to DIMINISHED.

1 b2	b3	4 b5	b6	b7	8
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THERE ARE 7 DIFFERENT SCALES THAT ALL CONTAIN THE SAME NOTES!

Let's look at our example again in the KEY OF G. Set your SLIDERULE up by putting G in the upper left window ...root note of the IONIAN MODE. There are 6 other scales that contain the same notes as a G MAJOR SCALE. They are ...

G	IONIAN	(G MAJOR DIATONIC)
A	DORIAN	
B	PHRYGIAN	
C	LYDIAN	
D	MIXOLYDIAN	
(E NATURAL MINOR)	E	AEOLIAN
	F#	LOCRIAN

Not only is this an interesting fact about how scales interrelate with one another, this also tells us everything about why we keep seeing those same **CHORD COMBINATIONS** over and over.

NOTICE THAT EACH OF THESE MODES HAS A CHORD EMBEDDED INSIDE IT.

In our example using the **KEY OF G**, we again see that familiar set of chords. Each chord is rooted on one of the **DEGREES OF THE MAJOR DIATONIC SCALE** and is represented by a **ROMAN NUMERAL**. Some of these chords are **MAJOR** and some are **MINOR** depending on whether the mode it is in has a **MAJOR 3rd** or a **MINOR 3rd** in it. These modes are always in the same order and retain their same major/minor orientation no matter which **KEY** you play them in so you absolutely have to memorize this set of relationships.

THIS ONE SYSTEM OF PITCH ORGANIZATION MAY EXPLAIN THE MELODIES AND CHORD PROGRESSIONS FOR 50-75% OF ALL THE SONGS YOU HAVE EVER HEARD!

I	G	2	B 4	D	6	7 8	G MAJOR is the TONIC CHORD in the key of G and can be found inside G IONIAN. But you're not playing this scale over the other chords in the song.
ii	A	2 C	4	E	6 b7	8	A MINOR is the ii CHORD in the key of G. It is embedded inside the Dorian Mode and A DORIAN is the scale you are playing to it.
iii	B b2	D	4	F# b6	b7	8	B MINOR is the iii CHORD. You'll be using B PHRYGIAN to jam with over this chord
IV	C	2	E	#4 G	6	7 8	C MAJOR is the IV CHORD. The 7 notes are functioning as C LYDIAN over this chord.
V	D	2	F# 4	A	6 b7	8	D MAJOR is the all important V CHORD and so MIXOLYDIAN is a common and most important scale.
Vi	E	2 G	4	B b6	b7	8	E MINOR is the Vi CHORD. It is the next most common chord after I-IV-V so AEOLIAN, the RELATIVE MINOR mode, is very common.
Vii	F# b2	A	4 C	b6	b7	8	F# DIMINISHED is the Vii CHORD. This chord has us playing LOCRIAN MODE over it. Note the b3 and b5 which make this scale a HALF DIMINISHED scale.

IF THESE SCALES ALL CONTAIN THE SAME NOTES AS A G MAJOR SCALE, WHAT'S THE HARM IN THINKING THAT YOU'RE PLAYING G MAJOR OVER ALL THESE CHORDS? WHY DO I NEED TO LEARN ALL THESE NEW SCALES WITH THESE WEIRD GREEK NAMES?

You see it's all about **FUNCTION - FUNCTION - FUNCTION!** Yes, these 7 notes will sound pretty good over these 7 chords but the more you listen, the more you begin to hear that each individual note sounds different depending on which chord you play it over. That is because each note changes **NUMERIC FUNCTION** with each chord change. Sometimes it is an **IMPORTANT** note like the 3 or the 5 and you will want to use it often ...hold it longer ...begin and end **PHRASES** with it. But over another chord, this same note might be functioning as a less important melody note like 4 or 6 and you will need to be more careful with it. Understanding these shifting relationships and being able to control them through **EAR TRAINING** is what separates the great improvisers from the "LICK BARFERS" and finger players. Yes it will take more work to learn to jam this way but the rewards are very real. You could learn to solo as well or better than many of your great guitar heroes!

THE MODE JAM TRACKS

LESSON 19

Hopefully you now understand the basic relationship between MELODY, SCALES and CHORD PROGRESSIONS in conventional DIATONIC MUSIC. You will find that there is a lot more to know as well but you've got a lot of it right here. These relationships are all laid out very nicely on your MUSICAL SLIDERULE and you can investigate them in any one of the 12 KEYS. Now it's time to begin mastering these relationships. You have a lot of FINGER PATTERNS to memorize and a lot of EAR TRAINING to do. It is going to take most of you YEARS to actually accomplish this. But don't let that put you off. All real musicians have to go through this and if you are destined to be a REAL PLAYER, you will not find the process too difficult or boring. By all means, take it at your own pace

Scotty's famous COLOR CODED FINGER CHARTS are awesome tools to help you learn your scales and modes. They allow you to see all the DEGREES OF THE SCALE at a glance and where all the octave notes are as well. They may seem a bit intimidating at first ...all these pages with all these patterns on each one. But it's not as bad as it looks. Over time you will notice a lot of REDUNDANCY built into these finger patterns. For example ...

THERE ARE 7 SCALES THAT ALL HAVE THE SAME NOTES IN THEM SO THE SAME FINGERING CAN BE USED TO PLAY ALL 7 SCALES!

Look at the charts for your 7 DIATONIC MODES ...IONIAN, DORIAN, PHRYGIAN, LYDIAN, MIXOLYDIAN, AEOLIAN and LOCRIAN. Notice that all the dots are actually in the same place on each page. The finger patterns you use to play these scales are all the same so you don't have 12 new shapes to master on each page. The only thing that changes from chart to chart is the COLOR of the dots indicating that the NUMERIC/HARMONIC FUNCTION of each note changes depending on which mode you are playing. A different note is functioning as the ROOT NOTE in each mode. Notice each of the finger boxes has a number. Lets compare some of them. Lets look at

IONIAN 1

Notice that the dots are all in the same place on each of these charts. RED is the ROOT NOTE and it's octaves. IONIAN is the 1st MODE and in this box it's root note is the lowest note you are playing.

DORIAN 1

Though the dots are all in the same place, the COLORS have all shifted up by 1 SCALE DEGREE.

Since DORIAN is the 2nd MODE, the ROOT (red) has moved up to what was the 2nd note of the previous mode, IONIAN.

PHRYGIAN 1

The same fingering can be used to play PHRYGIAN. It is the 3rd MODE. The dot colors have all shifted up by one more scale degree from the DORIAN and 2 up from the IONIAN.

It is rooted on the 3rd DEGREE of IONIAN MODE.

CHECK OUT YOUR OTHER MODE FINGER CHARTS AND YOU WILL SEE THIS PROCESS CONTINUING.

Here's another bit of redundancy to note. You know that all your strings are tuned 4 NOTES APART except G and B. So a scale fingering should remain the same when you move it over by a string as long as the ROOT NOTE IS UNDER THE SAME FINGER. The pattern will only change when you encounter the tuning difference between your G and B string. Notice how in these 3 examples, the root is always under your 2nd finger but on different strings. The finger patterns are then identical until you hit the B string. This is just one example of many ...

Ionian 3

Here is the 1st full octave of a MAJOR SCALE from the box IONIAN 3. It requires no finger shifts. Notice the familiar MAJOR SCALE interval pattern. The ROOT NOTE is under your 2nd finger on your low E string.

Ionian 8

In IONIAN 8, notice that the root note is also under your 2nd finger but it has moved over to be on your A string. Since this is also a MAJOR SCALE it retains the same interval pattern and so the same finger shape as IONIAN 3 just moved over one string.

Ionian 1

In IONIAN 1 the root is still under your 2nd finger but now it has moved to your D string. We see the same fingering again until we get to the B string. These last 3 notes must be shifted up a fret to adjust for the tuning.

Pity the poor guitar student who must slave away hour after hour with his boring old black and white finger charts and no real sense of what these scale notes sound like MELODICALLY over the chords in a song. Your MODE JAM TRACKS provide you with a RHYTHM SECTION (Bass, Drums and Rhythm Guitar) to IMPROVISE over anytime you want. Start by using it to simply make memorizing your scale fingerings less of a solitary pursuit. You get to hear how each note relates to each chord you play it over. This is EAR TRAINING. If you can stand it, try singing and saying the number of each note as you play it until you can sing the scale independent of hearing it on your guitar. Remember ...you don't solo with your FINGERS ...you solo with your MIND and your EARS! Eventually you will want to find a way to create your own jam tracks by acquiring recording gear of some sort. Put your drum machine on and strum the chords in the solo section of any song you want to learn to jam over or get the guys in the band to play that section over and over while you sit out and record it. Don't waste rehearsal time. It often takes a while to develop good melodic ideas from your scales. Take that recording home and jam your brains out until you get it right!

But for now, these tracks will be a good place to start. They contain several 2 chord progressions utilizing the most common changes in DIATONIC MUSIC. There is also a BLUES PROGRESSION that you can jam to a little later after we learn about BLUES HARMONY. We recorded these progressions all in the KEY OF A but you will want to learn to apply this way of thinking to all 12 keys. You have a lot of patterns to master particularly if you ever want to play JAZZ.

Start by cutting your scale charts into FLASH CARDS. This way you can choose any you want and put them right next to each other for visual comparison. You might look first at the ones with no FINGER SHIFTS (boxes 3 and 10) then 1 shift (box 5) then 2 shifts and so on. This may seem like a lot more work than you were expecting but the more fingerings you know for a scale, the more FREEDOM you have to get your melodic ideas out of your head and on to your fretboard quickly and smoothly.

Different mode types with the same BOX NUMBER will have the same finger pattern. For example, the dots on IONIAN 3 and LYDIAN 3 will be positioned in the same place but their color coding will be different showing that they have a MODAL RELATIONSHIP to one another. So to explore your mode fingerings, you will be choosing LIKE BOX NUMBERS to play over your mode jam progressions. Here is the key to the progressions included in your MODE JAM TRACKS....

TUNING NOTES		If your ELECTRONIC TUNER tells you that these tuning notes are sharp or flat by say 10 cents, tune your guitar accordingly.
I PROGRESSION 1 IV A MAJOR - D MAJOR		A IONIAN TO D LYDIAN A very common progression. Grab an IONIAN and a LYDIAN flash card with the same BOX NUMBER. Locate the place on the neck where the RED DOTS on the IONIAN box will be A's. Eventually try to memorize as many boxes as you can.
I PROGRESSION 2 V A MAJOR - E MAJOR		A IONIAN TO E MIXOLYDIAN Grab an IONIAN and a MIXOLYDIAN card with the same box number. Combine this thinking with the previous progression and you've got your I-IV-V PROGRESSION. Again find where you have to be to play in the key of A. Try all the boxes.
I PROGRESSION 3 Vi A MAJOR - F# MINOR		A IONIAN TO F# AEOLIAN A MAJOR CHORD and a MINOR CHORD in that all important RELATIVE MINOR relationship. The Vi CHORD is probably the next most common after I-IV-V making this a very common and important scale/chord relationship to master.
Vi PROGRESSION 4 ii F# MINOR - B MINOR		F# AEOLIAN TO B DORIAN When we learn about MINOR KEYS, you will find that this is better thought of as a I-IV PROGRESSION in the key of F# MINOR. Since we are working off the AEOLIAN MODE make sure your ROOT NOTE IS F# INSTEAD OF A.
I IV V PROGRESSION 5 A-D-E 12 BAR BLUES		A MINOR PENTATONIC BLUES MUSIC is different than normal DIATONIC MUSIC, We will be talking about it in a few more lessons. At that time, you will find this progression useful to jam over with an A MINOR PENTATONIC SCALE.

You will come to find out that these scale/chord relationships are not carved in stone. The really clever players can use ANY NOTE OVER ANY CHORD IN ANY KEY AT ANY TIME. But this requires very well trained ears and we approach this level of playing by first learning how to integrate this more conventional thinking.

OTHER SCALES, OTHER MODES LESSON 20

You probably know that there are many more scales in the world than those that we have looked at so far. This program is geared towards people mostly interested in playing everyday AMERICAN POP/ROCK MUSIC. Those of you who belong to various ethnic groups might have grown up listening to music that utilizes scales that sound quite different from the ones on our MUST KNOW LIST. I am not an ETHNOMUSICOLOGIST and it is beyond the scope of this program to address the music of other cultures. I have included a list of EXOTIC SCALES that I have gathered over the years from various sources and you may find it interesting to play and listen to them. Some of them do wind up in Western Pop on occasion and you will recognize them. Others can be found in more advanced WESTERN FINE ART MUSIC and the experimental music of the AVANT-GARDE.

Most of you will first want to concentrate on the MUST KNOW LIST. But there is an expanded list of scales that do show up in more conventional Western Music and we will look at them here.

Many people know about the system of MODES that we have been talking about for the last few lessons. But most people don't know that the system of modes based on the MAJOR DIATONIC SCALE is not the only set of modes that there are. Indeed, ANY SCALE can generate a system of modes. Often they aren't worth investigating (as in the case of the SYMMETRICAL SCALES) because no new and useful scale patterns will be discovered. But there are a few scales on our list that will generate meaningful systems of modes.

Consider the MELODIC MINOR SCALE. It has a system of 7 modes, several of which can show up in pop/rock music...not so much in ordinary DIATONIC chord progressions ...but in odd or advanced progressions containing MODULATIONS (where the song actually changes key) or CHORD SUBSTITUTIONS. Two of these scales, LYDIAN b7 and MIXOLYDIAN b13, are actually quite common and after you have gained a fair proficiency with the DIATONIC and PENTATONIC SCALES, you might want to turn your attention to these. There are finger charts for all 7 MELODIC MINOR MODES in your support material. Notice that they only differ from your familiar DIATONIC MODES by one note and are named accordingly.

START WITH 2 OCTAVES OF A MELODIC MINOR SCALE.



1 2 b3 4 5 6 7 8 MELODIC MINOR

1 b2 b3 4 5 6 b7 8 DORIAN b9

1 2 3 #4 #5 6 7 8 LYDIAN #5

1 2 3 #4 5 6 b7 8 LYDIAN b7

MIXOLYDIAN b13 1 2 3 4 5 b6 b7 8

LOCRIAN b9 1 2 b3 4 b5 b6 b7 8

(Notice the b4=3) **ALTERED 1 b2 b3 b4 b5 b6 b7 8**

You may see these scales labeled differently by different publishers. For example, DORIAN b9 may be called DORIAN b2 etc. and since each of these scales is close to 2 DIATONIC MODES, LYDIAN b7 could also be called MIXOLYDIAN #4 etc.

EXOTIC SCALES

1	b2	b3	4	5	b6	7	8			
1	b2	b3	4	5	6	7	8			
1	b2		3	4	b5	6	b7	8		
1	b2		3	4	5	b6	7	8		
1	b2		3	#4	#5	#6	7	8		
1	2	b3		5	b6		8			
1	2	b3		#4	5	b6	7	8		
1		#2	3	#4	5	6	b7	8		
1	b2		4	5	b6		8			
1	b2		4	b5		b7	8			
1	2	3	4	5	b6	b7	8			
1	b2	b3	3	4	b5	b6	b7	7	8	
1	b2	b3		5		b7	8			
1	2	b3		#4	5	b6	b7	8		
1	b2		3	4	5	b6	b7	8		
1	2	3	4	b5	b6	b7	8			
1	2	3		#4	5	b6	b7	8		
1	2	3		#4	5	6	b7	8		
1	2	3		#4	#5	#6	7	8		
1	2	3	4	b5	b6	b7	8			
1	b2	b3		5	b6		8			
1	b2		3	4	5	b6	7	8		
1	2	b3	4	5	b6	7	8			
1	b2	b3	4	5	6	b7	8			
1	b2		3	4	b5	b6	7	8		
1	2	b3		#4	5	b6	7	8		
1	b2		3	4	5	b6	7	8		
1	2	b3	4	5	6	7	8			
1	b2		3	4	5	b6	b7	8		
1	2	3		5	6		8			
1	2	b3	3	4	5	b6	6	b7	7	8
1	b2		3	4	5	b6	b7	8		
1	2		4	5		b7	8			
1	b2		4	5	b6		8			
1	2	3		5	6		8			
1		3	#4	5		7	8			
1	2	3	4	5	b6	7	8			

Neapolitan Minor
Neapolitan Major
Oriental
Double Harmonic Major
Enigmatic
Hirajoshi
Hungarian Minor
Hungarian Major
Kumoi
Iwato
Hindu
Spanish 8 Tone
Pelog
Hungarian Gypsy
Major Phrygian
Major Locrian
Lydian Minor
Overtone
Leading Whole Tone
Arabian
Balinese
Gypsy
Mohammedan
Javanese
Persian
Algerian
Byzantine
Hawaiian
Jewish
Mongolian
Ethiopian
Spanish
Egyptian
Japanese
Chinese
Chinese 2
Harmonic Major

PENTATONIC MODES

LESSON 21

PENTATONIC SCALES are some of the best known scales in popular music. The MAJOR and MINOR PENTATONIC SCALES are on our MUST KNOW LIST. PENTATONIC means 5 NOTES. These scales are said to have 5 notes. But I will encourage you to include the OCTAVE note and so they actually have 6 notes. Since these scales have fewer notes than the DIATONIC SCALES, their notes are separated by LARGER INTERVALS. Pentatonic Scales contain no half steps (minor 2nds) and each contains two 1 and 1/2 step intervals (augmented 2nds). As a result, these scales have an open and uncluttered sound that gives them many special properties. They can actually function as several different DIATONIC SCALES at the same time. For example, since a MAJOR PENTATONIC SCALE has no 4th or 7th, it could function as a simplified version of the IONIAN, LYDIAN or MIXOLYDIAN MODES since the 4 and 7 are the only notes these scales don't share.

Our MUST KNOW LIST contains only two PENTATONIC SCALES ...MAJOR and MINOR. But many people don't know that PENTATONIC SCALES HAVE MODES. There are in fact many Pentatonic Scales. How do we know this? Remember when we first learned about DIATONIC MODES? We talked about how many people first learn to improvise. We looked at that argument between the three guys ...

WOW! I CAN PLAY ONE SCALE OVER SEVERAL CHORDS!

NO! WHEN THE CHORD CHANGES, SO MUST THE SCALE CHANGE!

THIS IS THE THING ABOUT MODES. THE SAME SET OF NOTES IS SEVERAL DIFFERENT SCALES. THE QUESTION IS WHICH NOTE IN THE SET IS "DO"?

We first looked at this within the context of DIATONIC HARMONY involving the modes of the MAJOR DIATONIC SCALE. But often, our first experience with scales in improvisation involves the PENTATONIC SCALES. The Pentatonic Scale (or Minor Pentatonic) is often the first scale we are shown as fledgling Rock guitarists. You might have thought you were playing a Pentatonic Scale over the multiple chords in a progression, but in fact, as in the case of the Diatonic Modes, you were actually playing various MODES of the PENTATONIC SCALE over these chords. Again, this is significant because the NUMERIC/HARMONIC FUNCTION of each note will change depending on which chord you are playing it over. In most contexts, we are using these Pentatonic Modes in the same way we used the Diatonic Modes ...over the same types of progressions. It's just that the Pentatonic Modes contain TWO FEWER NOTES. At first, this seems to make these modes easier to control ...less notes to worry about. For example, the 4 and the 7, notes that are left out of a Pentatonic Scale, are notes that you have to be more careful with when constructing melodies. They won't sound as good as the 1st, 3rd and 5th, notes that are in the chord itself. So you may not want to start or end melodic phrases with them. You may not want to dwell on them as long as other notes. Remember, when using a note in a melody you not only have to consider it's PITCH but also it's DURATION. Notes that add spice and interest when used sparingly can actually sound like wrong notes when you hang on them too long. Once your ears are trained, you will be able to hear these relationships more clearly and you will notice that a lot of players play a lot of dumb notes because they aren't really listening.

Some players believe that the solution is to simply leave these notes out and just play Pentatonic Scales. But I'm not so sure. Why would you want only 6 notes in your arsenal when you could have 8? That's 25% of the notes you could use to make your melodies 25% more interesting! Better we should train our ears and learn to control these difficult notes rather than just leaving them out. In fact, what you will come to see is that the really clever players can use ANY OF THE 12 NOTES OVER ANY CHORD IN ANY KEY ANYTIME! So you may not want to be a Pentatonic player ...you may not even want to be a Diatonic player. I think that all players who really understand music are trying to become true CHROMATIC PLAYERS. This doesn't mean that you try to cram every last note into every little space and OVERPLAY (as you will hear some players do). It just means that you have more choices at any given moment and that's a good thing. In fact many of the people telling you that Pentatonic Scales are so significant don't really understand enough about what they're doing to have an intelligent opinion. There are a lot of "meat heads" in Rock guitar. They'll tell you that they use Pentatonic Scales but if you go back and analyze what they just played, you'll find that they actually were using notes outside of the Pentatonic Scale without even knowing it!

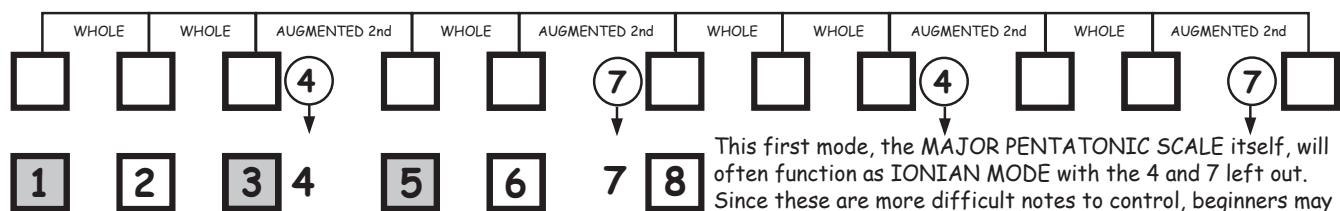
There may have been some justification for leaving out the 4 and 7 over the I chord in the progression, but as you investigate the PENTATONIC MODES over the other chords in the progression, you will see that you are leaving out more useful notes like the 3rd and the 5th. In fact, over the IV chord and the VII chord, you will be leaving out the ROOT NOTE! How dumb is that?

PENTATONIC SCALES HAVE MANY USES BUT YOU WILL MOST OFTEN SEE THEM AS SIMPLIFIED VERSIONS OF DIATONIC MODES

That is they work over normal DIATONIC PROGRESSIONS but each mode has 2 fewer notes. Sometimes these are the more difficult notes to control but not always. As you go through the PENTATONIC MODES you will see yourself missing some of your most useful and powerful notes.

START WITH 2 OCTAVES OF A MAJOR PENTATONIC SCALE.

IT'S LIKE A MAJOR DIATONIC SCALE WITH THE 4 AND 7 LEFT OUT. NOTICE THE TWO AUGMENTED 2nd INTERVALS.



This is the 4th mode, PENTATONIC LYDIAN. Played over the IV chord, this is a common mode.



PENTATONIC MIXOLYDIAN is your 2nd most common mode since it is played over the V chord. Once again we skip important notes



Recognize this? The MINOR PENTATONIC SCALE is the 6th mode of the MAJOR PENTATONIC SCALE! Very important to know. It's that RELATIVE MINOR thing.



In PENTATONIC LOCRIAN, we once again wind up leaving out the ROOT NOTE. Though DIMINISHED CHORDS are much less common in Pop Music, when they do show up, it would be pretty dumb to leave out b5.



TAKE A LOOK AT YOUR MAJOR AND MINOR PENTATONIC FINGER CHARTS.

Since these two scales are MODES of each other (the Minor Pentatonic is the 6th mode of the Major Pentatonic), the same fingerings can be used to play either scale. Notice that the ROOT of the Minor Pentatonic (red dot) is the 6th (indigo dot) of the corresponding Major Pentatonic box. Many players don't know whether they're using these fingerings as Major or Minor at any given moment. The question is are you playing it over a Major or Minor chord or chord progression?

Minor Pentatonic Scales are very big in BLUES and BLUES ROCK MUSIC and we will be looking at them more in depth in a few lessons when we look at these styles of music.

I don't mean to sound like I am discouraging you from taking Pentatonic Scales seriously. Their uncluttered sound makes them very useful in many contexts. It's just that too many players who don't know their theory and haven't trained their ears miss a lot of opportunities when playing strictly Pentatonically over conventional progressions. I myself am striving to learn to control more of the 12 notes ...not less! Many players play that way because they just don't know any better.

MUSIC IN MINOR KEYS

LESSON 22

THIS IS HUGE! HOW MANY KEYS ARE THERE TO PLAY MUSIC IN?

Up to now we have been saying that there are 12. But those are just the MAJOR KEYS ...based on the MAJOR DIATONIC SCALE and it's MODES and embedded TRIADS. There are also 12 MINOR KEYS to play in. Perhaps half of our songs are in these MINOR KEYS. How do you tell if a song is in a MINOR KEY?

How do you generally tell what key any song is in? As I have said before, this is a complex issue with a number of different facets. You kind of develop evidence towards a particular conclusion and when enough of these indicators point toward one answer, you make the judgment as to what key you are in. What is the FIRST CHORD in the song? What is the LAST CHORD in the song? What KEY SIGNATURE is printed in the sheet music? None of these alone is guaranteed to give you the absolute answer but they are all helpful.

I always ask myself which chord feels like the "HOME CHORD". Which chord begins and ends the most PHRASES? Which chord feels like the point from which everything else flows and then ultimately returns to? If that chord is a MAJOR CHORD, you are probably in a MAJOR KEY. If that chord is a MINOR CHORD, you may well be in a MINOR KEY (although in both cases you might also be in a MODAL KEY which we will discuss in our next lesson). You may have trouble hearing which chord is the HOME CHORD at first and so this becomes one of your earliest EAR TRAINING challenges. Since Minor Chords sound SAD, songs in a MINOR KEY are apt to be the more melancholy songs you hear on the radio.

We spent several hours learning about MAJOR DIATONIC HARMONIZATIONS (Major Key modes and progressions). Since MINOR KEY music is just as big, you might think that it will take hours to learn about them as well. But the good news is ...if you understand MAJOR DIATONIC HARMONY, you already know most of what there is to know about MINOR KEYS. In fact, I'm fond of saying that the whole subject can be summed up in three simple words ...

Vi BECOMES i

SIX BECOMES ONE. The 6th DEGREE OF THE MAJOR DIATONIC SCALE becomes the TONIC for the MINOR KEY. This is the main and true meaning of the concept of RELATIVE MINOR. Every MAJOR KEY is said to have it's RELATIVE MINOR KEY rooted on it's 6th DEGREE. So, for example, the key of A MINOR is the RELATIVE MINOR KEY of the key of C MAJOR because A is the 6th of C. The RELATIVE MINOR of G MAJOR is E MINOR. The RELATIVE MINOR of Bb MAJOR is G MINOR etc. All these relationships are laid out on your MUSICAL SLIDERULE. The MAJOR KEY and it's RELATIVE MINOR KEY share the SAME KEY SIGNATURE since they both have the same notes in them.

MAJOR KEY music involves HARMONIZING a MAJOR DIATONIC SCALE. As we learned in our previous few lessons, all our basic SCALE / CHORD PROGRESSION / MELODY relationships are obtained by looking at the MODES of the MAJOR DIATONIC SCALE and noticing that each of these modes has a chord embedded inside it.

The 6th mode of the Major Scale is the AEOLIAN MODE. In this context it is more often referred to by it's other name, THE NATURAL MINOR SCALE. To understand the SCALE / CHORD PROGRESSION / MELODY relationships in MINOR KEY MUSIC, we are now going to learn to HARMONIZE the NATURAL MINOR SCALE and observe it's MODES and embedded TRIADS. Since the AEOLIAN is one of the modes of the IONIAN, we are not going to encounter any new scales here. We will see our old familiar DIATONIC MODES simply starting this time with AEOLIAN. They are even in the same order as they were in the Major Key. Look at your SLIDERULE. Notice that each of your DIATONIC MODES has a ROMAN NUMERAL in front of it. By now you should know all about these (the DEGREES of a MAJOR DIATONIC HARMONIZATION). Now notice that there is a second Roman Numeral at the END of each mode. These correspond to the DEGREES of the NATURAL MINOR SCALE. Notice that AEOLIAN is now mode #1. All the other modes fall right into line. Locrian is now mode #2. IONIAN (the TONIC MODE in a Major Key) is now the 3rd mode etc. As always, there are chords embedded in these modes.

THIS SHIFT GENERATES A WHOLE NEW TYPE OF CHORD PROGRESSION.

- i MINOR - (AEOLIAN) NATURAL MINOR
- ii DIMINISHED - (LOCRIAN)
- III MAJOR - (IONIAN)
- iV MINOR - (DORIAN)
- V MINOR - (PHRYGIAN)
- VI MAJOR - (LYDIAN)
- VII MAJOR - (MIXOLYDIAN)

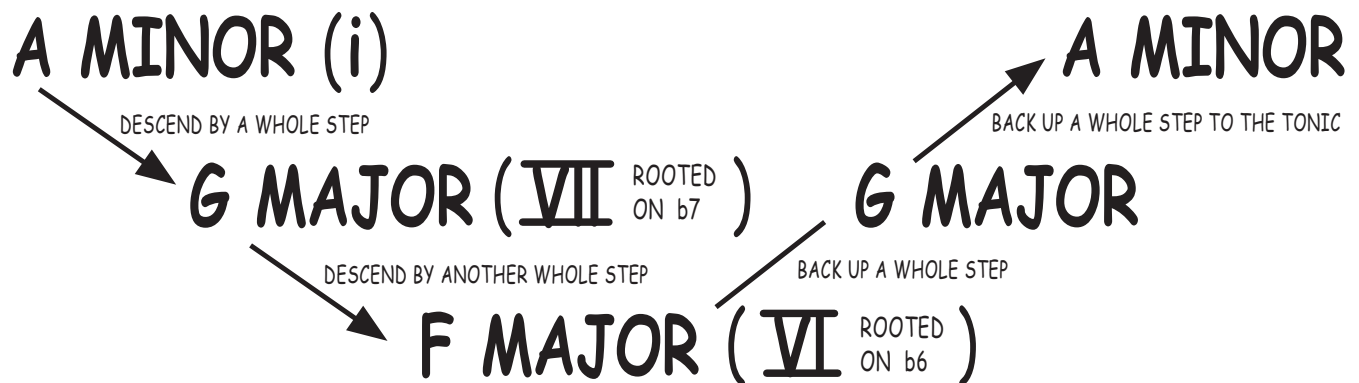
LET'S LOOK AT 2 OCTAVES OF A NATURAL MINOR SCALE (AEOLIAN)

	WHOLE	HALF	WHOLE	WHOLE	HALF	WHOLE	WHOLE	WHOLE	HALF	WHOLE	WHOLE	HALF	WHOLE	WHOLE
i	1	2 b3	4	5 b6	b7	8	The first mode in a MINOR KEY is AEOLIAN, the NATURAL MINOR SCALE itself. So the HOME CHORD is a MINOR CHORD. As a result, MINOR KEY MUSIC sounds SAD.							
ii ^ø	1 b2	b3	4 b5	b6	b7	8	The 2nd mode is LOCRIAN. It is a HALF DIMINISHED SCALE. A FULL DIMINISHED SCALE would contain a bb7 (which is actually a 6).							
III	1	2	3 4	5	6	7 8	IONIAN is the 3rd mode but notice it is rooted on the b3 of the NATURAL MINOR SCALE. It is the TONIC MODE in the RELATIVE MAJOR KEY.							
iV	1	2 b3	4	5	6 b7	8	The 4th mode is the old 2nd mode, DORIAN. So the iV CHORD in a MINOR KEY is a MINOR CHORD.							
V	1 b2	b3	4	5 b6	b7	8	PHRYGIAN is the 5th mode so the V CHORD is also a MINOR CHORD.							
VI	1	2	3	#4 5	6	7 8	LYDIAN, the 6th mode is rooted on b6 of the NATURAL MINOR SCALE so the VI CHORD is a MAJOR CHORD.							
VII	1	2	3 4	5	6 b7	8	The 7th mode, MIXOLYDIAN is rooted on b7 so the VII CHORD is MAJOR, rooted a WHOLE STEP below the TONIC in MINOR KEY MUSIC.							

SO IN MINOR KEYS, i-iV-V ARE ALL MINOR CHORDS !

THIS IS BECAUSE, IN A MAJOR KEY, THE SPACING OF I-IV-V IS THE SAME AS THE SPACING OF Vi-ii-iii. So when Vi becomes i, the old ii and iii (Dorian and Phrygian) become the new iV and V in the RELATIVE MINOR KEY.

Another classic MINOR KEY CHORD PROGRESSION involves STARTING ON THE i CHORD (MINOR), DESCENDING by a WHOLE STEP to the VII CHORD (MAJOR) and then DOWN another WHOLE STEP to the VI CHORD (MAJOR) and then back UP AGAIN. How many hundreds of songs have you heard like this? Try this example in the key of A MINOR ...



THERE IS STILL A BIT MORE TO KNOW ABOUT MINOR KEYS AND WE WILL LOOK AT THAT IN OUR NEXT LESSON.

MODAL MUSIC

LESSON 23

Actually, we'll talk about MODAL MUSIC in just a minute.. There is still one more BIG thing to know about MINOR KEY MUSIC.

REMEMBER WE SAID THAT A

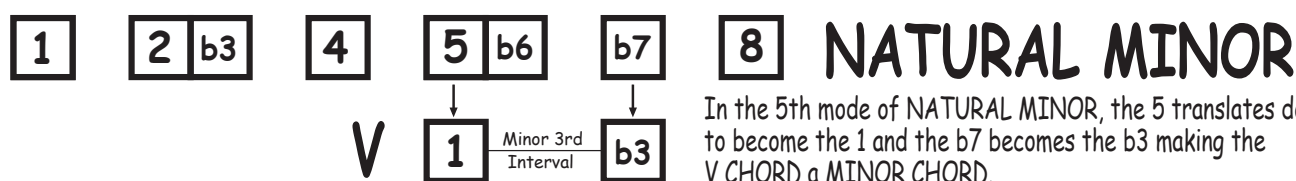
i-iv-v PROGRESSION IN A MINOR KEY CONTAINS ALL MINOR CHORDS.

WELL THAT ISN'T ALWAYS TRUE. IN FACT, MORE OFTEN THEN NOT,

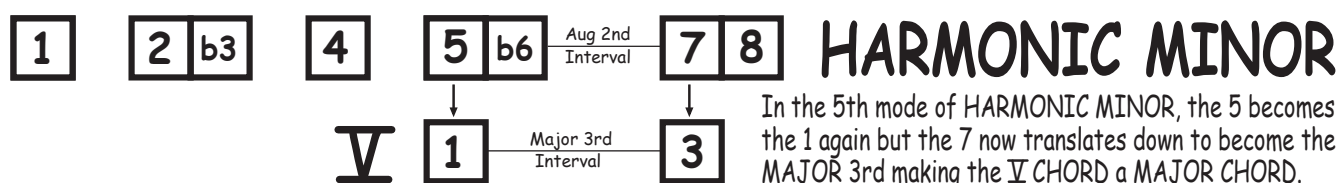
**THE V CHORD CAN BE A MAJOR CHORD
THE SAME WAY IT IS IN MAJOR KEY MUSIC !**

That's why we created the **HARMONIC MINOR SCALE**.

There is yet another HARMONIZATION that you should know about. HARMONIC MINOR is not a scale that you PLAY that often. It is more a THEORETICAL CONSTRUCT that justifies a certain type of CHORD PROGRESSION. If we look at the MODES and EMBEDDED TRIADS that occur when we HARMONIZE a HARMONIC MINOR SCALE, we see that the V CHORD becomes a MAJOR CHORD. What is the only difference between NATURAL MINOR and HARMONIC MINOR? Notice that in HARMONIC MINOR, the 7th has been raised by a half step up from b7 to 7. (This also opens up that AUGMENTED 2nd INTERVAL between the b6 and the 7.) In a NATURAL MINOR HARMONIZATION, the b7 would translate down to become the b3 in the 5th MODE causing the V CHORD to be a MINOR CHORD. By shifting the b7 up to 7 in HARMONIC MINOR, we shift the b3 in the 5th MODE up to become a MAJOR 3rd, changing the V CHORD INTO A MAJOR CHORD!



In the 5th mode of NATURAL MINOR, the 5 translates down to become the 1 and the b7 becomes the b3 making the V CHORD a MINOR CHORD.



In the 5th mode of HARMONIC MINOR, the 5 becomes the 1 again but the 7 now translates down to become the MAJOR 3rd making the V CHORD a MAJOR CHORD.

Of course the 5th is not the only mode effected by this shifted 7th. All 7 modes are altered and two other EMBEDDED TRIADS also change. The 3rd mode changes from major to augmented and the 7th mode changes from major to diminished. This harmonization is illustrated on your sliderule and also in Lesson 30. In my experience, there isn't a lot of popular music that utilizes this harmonization. Most use the NATURAL MINOR harmonization. But be prepared for the V chord to be either major or minor.

AN IMPORTANT QUESTION THAT I DID NOT ADDRESS IN THE VIDEO

So what SCALE might you use to create melodies over a MAJOR V CHORD in a MINOR KEY SONG?

You wouldn't use PHRYGIAN as you would in a NATURAL MINOR harmonization because it is a MINOR SCALE.

You could raise the b3 up to a 3 turning it into a scale called PHRYGIAN MAJOR. You can find this scale on your list of EXOTIC SCALES on page 20A. It is spelled 1-b2-3-4-5-b6-b7-8. The problem with this solution is that now you have an AUGMENTED 2nd interval between the b2 and the 3 and this can make the scale sound too "middle eastern" for our "western" ears.

So you might also raise the b2 up to a 2. This scale is then spelled 1-2-3-4-5-b6-b7-8. This scale is called MIXOLYDIAN b13. It is the 5th mode of the MELODIC MINOR SCALE. So now you have a "western" sounding major scale that retains the b6 and b7 from the original PHRYGIAN MODE. Since the V chord is so often MAJOR in MINOR KEY songs, this is an important scale to know.

MODAL MUSIC

SOME SONGS AREN'T IN MAJOR OR MINOR KEYS

EVEN THOUGH THEIR SCALE/CHORD PROGRESSION/MELODY RELATIONSHIPS STILL RELATE TO THE MAJOR SCALE.

Grab your SLIDERULE and consider this simple CHORD PROGRESSION. What KEY is it in?

Em - A

The PROGRESSION begins with Em. Which CHORD would you end with? Probably still Em. Which CHORD feels like the "HOME CHORD"? The answer is again Em. So you're thinking this PROGRESSION is in the KEY of E MINOR. So you start to jam using E AEOLIAN. But you notice there is a really bad wrong note in there. What's up with that? So you grab your SLIDERULE and set it up to be in Em (put E in the RELATIVE MINOR window in the Vi MODE, AEOLIAN and use the second set of ROMAN NUMERALS at the end of each MODE). Em is now the i CHORD and A is the iv CHORD. What's wrong with this picture?

IT CAN'T BE IN E MINOR BECAUSE A IS THE iv CHORD AND SHOULD BE MINOR.

So you think somehow you miscalculated. Maybe it's in the KEY OF A MAJOR. So you put A in the 1st IONIAN window at the upper left.

BUT IT'S NOT IN A MAJOR. E WOULD BE THE V CHORD AND SHOULD BE MAJOR.

So what KEY is it in? The KEY OF D MAJOR is the only KEY that has an Em and an A CHORD in it. So put D in the upper left window. Em is now the ii CHORD (DORIAN-MINOR) and A is the V CHORD (MIXOLYDIAN-MAJOR). So you begin jamming with a D MAJOR SCALE (you're actually playing E DORIAN and A MIXOLYDIAN) and everything sounds good. That wrong note is gone.

BUT IT'S NO GOOD THINKING YOU'RE IN D MAJOR IF THERE IS NO D ROOT IN THE SONG.

So what KEY are we in? Since Em feels like the HOME CHORD and is occupying the DORIAN position in this particular progression,

THIS IS A PIECE OF MODAL MUSIC SAID TO BE IN THE KEY OF E DORIAN.

Now play this very common PROGRESSION. The D sounds like the "HOME CHORD".

D - C - G - D

So you think maybe it's in the KEY OF D MAJOR. But when you begin jamming with a D MAJOR SCALE you hear that wicked wrong note again. What's wrong this time? You grab your trusty SLIDERULE and set it up in D MAJOR. But notice that there is no C at all in the KEY OF D. Then maybe you notice that G-C-D is actually I-IV-V in the KEY OF G MAJOR. So you begin jamming with a G MAJOR SCALE and once again the wrong note goes away.

BUT WE CAN'T BE IN G MAJOR BECAUSE G DOESN'T FEEL LIKE THE "HOME CHORD".

So what KEY are we in? Since D feels like the HOME CHORD and occupies the MIXOLYDIAN position in this particular KEY,

THIS IS A PIECE OF MODAL MUSIC IN THE KEY OF D MIXOLYDIAN.

DORIAN AND MIXOLYDIAN ARE BY FAR THE MOST COMMON MODAL HARMONIZATIONS AND SO THE ONLY ONES WE WILL INCLUDE HERE.

DORIAN HARMONIZATION

1	2	b3	4	5	6	b7	8	i MINOR
PHRYGIAN								
1	b2	b3	4	5	b6	b7	8	ii MINOR
LYDIAN								
1	2	3	#4	5	6	7	8	III MAJOR
MIXOLYDIAN								
1	2	3	4	5	6	b7	8	IV MAJOR
AEOLIAN								
V MINOR	1	2	b3	4	5	b6	b7	8
LOCRIAN								
Vi DIM.	1	b2	b3	4	b5	b6	b7	8
IONIAN								
VII MAJOR	1	2	3	4	5	6	7	8

MIXOLYDIAN HARMONIZATION

1	2	3	4	5	6	b7	8	I MAJOR
AEOLIAN								
1	2	b3	4	5	b6	b7	8	ii MINOR
LOCRIAN								
1	b2	b3	4	b5	b6	b7	8	iii DIM.
IONIAN								
IV MAJOR	1	2	3	4	5	6	7	8
DORIAN								
V MINOR	1	2	b3	4	5	6	b7	8
PHRYGIAN								
Vi MINOR	1	b2	b3	4	5	b6	b7	8
LYDIAN								
VII MAJOR	1	2	3	#4	5	6	7	8

What else is there to know about HARMONY IN WESTERN MUSIC? Some CHORD PROGRESSIONS CHANGE KEY right within the song. This is called MODULATION and we will be talking about that in LESSON 31. Some music is just so weird that it can't be analyzed in any conventional context. This can run the gamut from advanced EXPERIMENTAL MUSIC to FREE IMPROVISATION to know-nothings making a bunch of weird, ugly noise which, sadly, so much ROCK MUSIC has become.

But there is one more style of HARMONIZATION that is very popular and important to understand. That is this stuff we call THE BLUES. Let's look at it now.